

Chapter Title: FORAGING FOR A FIELD STATION CULTIVATING MATERIAL EFFICIENCY
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Book Title: Fabricate 2024

Book Subtitle: Creating Resourceful Futures

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Published by: UCL Press. (2024)

Stable URL: <https://www.jstor.org/stable/jj.11374766.6>

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LOCALISE
RECLAIM
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LOCALISE

SOURCING AND PERFORMANCE

FORAGING FOR A FIELD STATION CULTIVATING MATERIAL EFFICIENCY

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ARCHITECTURAL ASSOCIATION

Introduction

Within industrial timber production, it is widely accepted that as little as 50% of a standing tree makes it into a building component. This is partly an outcome of the process of harvesting, where the crown timber and branches are left behind in the forest.

This project deploys an innovative robotic application to enable the use of this foraged small-diameter roundwood (50-90mm, usually a by-product of timber harvesting) as the primary building component in a 100m² roof system for a field station at Hooke Park in Dorset, UK. The application for this zero-value timber promotes a radical rethinking of the timber supply chain and a strategic repositioning of advanced technologies within the production of timber architecture.

Carefully nested within the fabric of the industrial woodland at Hooke, the Field Station is a flexible-use shelter – providing a field base for research and teaching, shelter for our forester and for passing walkers, and an outdoor classroom for local forest schools, as well as being a base for informal gatherings, talks, and events on site (Fig. 2). In the longer term, the Field Station is imagined

as an observatory for monitoring change, as the neighbouring agricultural landscapes in view from the Field Station undergo a rewilding programme and as the forestry growth and harvesting evolve around it.

The project resulted in a lightweight and demountable roof frame structure that is assembled without the use of permanent mechanical or glue fixings. Designed for disassembly, the roof structure comprises two planes of chords that are held together by threaded rod, which forces all the diagonal roundwood branches into compression allowing the use of straightforward tenon connections.

A forest context

Hooke Park is the Dorset campus of the Architecture Association (AA). Set within its own working woodland, it is a laboratory for large-scale experiments in timber architecture and for research that places the forest at the centre of the work. Because Hooke Park processes its own timber and manages the woodland that supplies it, the academic and research programmes based there are uniquely positioned to interrogate and innovate across the timber supply chain, from sapling to building.





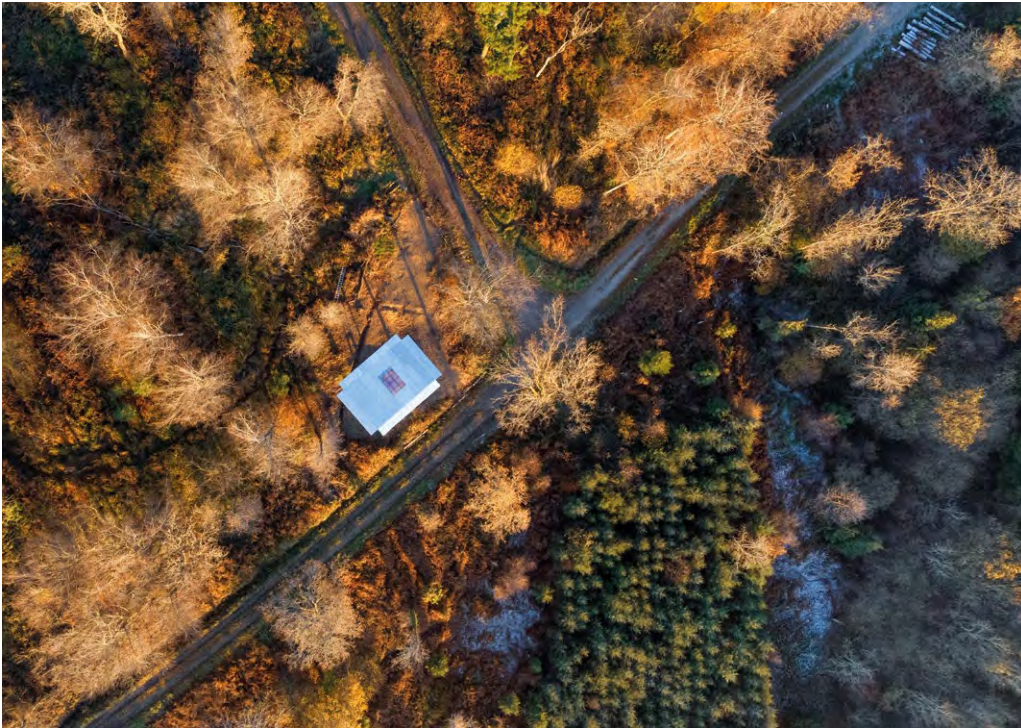
Situated apart from other campus buildings, and set within the working forest, the Field Station is a 100m² open air structure designed and built by students and staff from the AA's Design + Make postgraduate programme. (Fig. 1). The challenge was to devise an architectural application for 'forestry waste' crown timber – small-diameter roundwood of limited market value, commonly left on the forest floor as a by-product of the harvesting process – and to demonstrate how an undervalued resource might be transformed into valuable construction material.

The material parameters for the built demonstrators are in many ways defined by the silvicultural approaches and woodland management strategies of the past. Decision-making in forestry can be measured in decades if not centuries, and the project is envisaged as a space for students, researchers, and ecologists, as well as our forester, to work 'in the field' as they engage with the long-term thinking necessary for the development of future forest strategies for a climate-resilient, healthy woodland. Forest plantations for timber are not merely crops: they are complex ecosystems, increasingly understood as *intelligent* systems and akin to human societies, being complex, adaptive, and self-organising (Simard, 2021).

The Field Station uses two species of timber – Beech and Ash – both of which are at risk. Beech is among the leading British tree species considered most at threat from climate change (Hemery *et al.*, 2021, p.201), while Ash is under siege by a fungus that leads to dieback. Both tree species were harvested at Hooke in 2021 as part of the woodland management plan, and both provide material for the build. The forest residue – after the industrial harvester has passed through the Beech section – is made up of 50-90mm-diameter branch timber, which forms the primary building material deployed in the project. This material accounts for a substantial portion of timber left behind in the forest.

The structural logic was developed together with the engineering team at multinational consultancy Arup, resulting in a space truss that incorporates Beech roundwood braces within a dimensional Ash grid. This facilitates a 3m cantilever on all sides. The integration of a bespoke computer vision system with robotic fabrication ensured the precision of 256 roundwood braces, adapting to natural branch variations.

In this way, a relationship is somewhat inverted: the forest dictates material parameters, and these parameters lead the design, while technological strategies serve to elevate the status of a discarded material for an application of



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zero-value timber, working with what the forest at Hooke Park provides. It suggests an enhanced adaptive, localised approach to material resources.

The option of using small-diameter roundwood thinnings for the linear members was something the team was initially keen to explore. However, it found there was not enough timber of suitable length and diameter available for harvest from the forest at the time. Stitching shorter lengths together was a possibility, although it might well require a slightly different structural approach, and the top and bottom of these members being machined to achieve the accuracy. (The advantage of the double frame-sawn linear members is that they automatically generate the nodal connection.) This investigation simply proved unfeasible within the academic timeframe of the project. Although it was a trade-off, the explicit constraint set to students was to work only with what our forest could provide, and it was also important to hold to this ambition. The Ash used for the sawn-timber grid locating the diagonals, was sourced from our own forest. Since dieback has affected trees in Hooke Forest, some Ash is being harvested early (Figs. 5, 6).

Pursuing a wholly roundwood space-truss construction is a proposition that would be interesting to develop further, as it could be argued that the top and bottom linear

1. As a point of observation, the project engages with, and bears witness to, its immediate surroundings. © Design + Make Postgraduate Programme, Hooke Park.
2. The strategic location of the Field Station, firmly nested within the forest context. © Design + Make Postgraduate Programme, Hooke Park.
3. The industrial harvester at work in a Beech stand at Hooke Park. © Frederik Petersen.
4. The tenons and shoulders on robotically machined small-diameter roundwood. © Design + Make Postgraduate Programme, Hooke Park.

members as machined timber are dominant, and might feel counter to the objectives of the project. Yet there is something unexpected in the aesthetic contrast within the structure; the clean lines of the sawn timber are striking when seen alongside the natural branch forms, and perhaps this tells its own story.

The brief for the Field Station required that it touched the earth lightly, and that it was temporary and demountable, exemplifying principles of flexibility, reuse, and future reclamation, in keeping with ideas of the circular economy. The lack of permanent mechanical fasteners or glue, using only central tensioning rods at each node, which can be easily removed, facilitates demounting. The components were fabricated, flat-packed, and transported on site for a swift assembly process, completed in just ten days (Figs. 11, 12, 13, 14). This flexibility offers a dynamic shelter within an evolving working forest. The Field Station's mobility offers the potential for relatively simple relocation, capable of adjusting to the woodland's periodic thinning and felling over time. In its present position it faces south, framing a changing landscape – oriented towards a spruce compartment felled and replanted five years ago, and looking out towards the neighbouring Mapperton Estate, which is in the midst of a large-scale rewilding project. It serves as an observatory, a teaching space, a shelter for the forester, a base for researchers and ecologists measuring soil health, counting moths and bats, a resting point for walkers. It is a space for witnessing and monitoring change in the landscape over extended periods of time.

Crown timber, which is typically discarded as waste in contemporary forestry operations, is the key component and research focus for the project. Depending on the type and age of the tree, this material may account for 20-50% of its overall above-ground biomass. A 2015 global tree study (Crowther *et al.*, 2015) concluded that at that time Canada had 8953 trees per person, the US 716, France 182, and the UK just 47 trees per person. Given that in tree terms the UK is one of the poorest nations in the world, with woodland cover comprising only 13% of the country's total land area (Forest Research, 2023), it seems important to use as much as possible of a harvested tree, and for designers to create use cases towards this goal. By using branch wood braces rather than traditional dimensional lumber, the amount of timber saved was equivalent to two mature Beech trees. In a nation with little forest cover, this method significantly reduces the material waste of harvested trees from 50% to 10% (Fig. 3).

To forage the material for the build, students spent many hours searching the forest floor for Beech branches that

met the project's requirements for size and quality – a form of visual grading. This is reminiscent of earlier foraging traditions known as *estover*, a term that refers to an allowance of wood that a person can take from a commons, and derives from the French *estovoir*, meaning *that which is necessary* (Fig. 4).

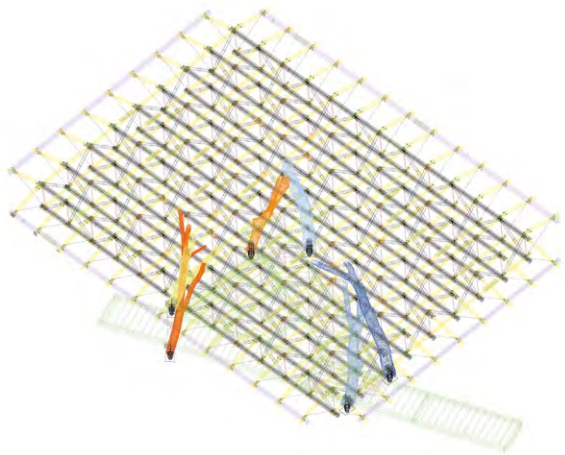
This project fits into a broader context within the Design + Make postgraduate programme at Hooke Park, which looks at minimally processed timber and aligns with a lineage of research established at Hooke Park in the 1980s by John Makepeace, Frei Otto, and Adrens Burton Koralek, which involved testing the structural applications of forestry by-products through constructing buildings on site. Over the past eight years, a series of projects – Boiler House, Woodland Cabin, and Woodchip Barn (Self and Vercruysse, 2017) – have explored the unique structural geometries of trees, starting at the ground, and now, with the Field Station, working up into the crown. Other projects at Hooke Park, such as the Foundry (Vercruysse *et al.*, 2019) and the library skeleton (Vercruysse, 2020a), have advanced the scope of robotic workflows in the built work (Vercruysse, 2020b). The innovation in the Field Station – and the departure from robotic workflows used on past projects – lies in the integration of the bespoke vision system.



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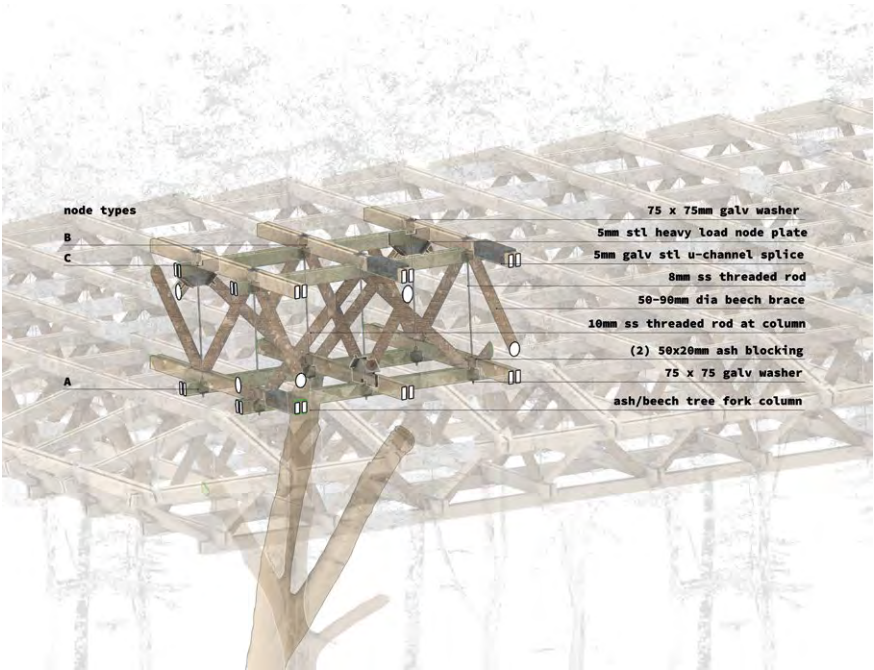


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A vision system

By integrating a vision system within the Hooke Park robotics cell, crown timber branches can be processed into precise, predictable geometries, accounting for natural variation in an element’s geometry. With every web element in the space-frame system having its own unique shape at the time of harvest, an automated workflow comprised of scanning, part analysis, and toolpath generation enabled an efficient production of structural components, processed into connection details only where the branch meets the building.

The guiding framework for this robotic setup was the delivery of 256 identical components from non-standard timber. The ambition was to build a robotic application to process leftover Beech crown timber into high-quality structural products. Within the design of the Field Station, the elements made from these branches fit into the building’s structural system by the geometry defined at their extremities. However, there are no geometric conflicts with any other elements along their length. By integrating a vision system with an indexing circular saw, non-uniform geometries could receive a series of targeted cuts at their ends, extracting a pair of in-plane tenon joints from the branch volume.

This application for robotic vision is enabled by an Ensenso N30 structured light scanner, typically used



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in factory settings for quality inspection or bin picking. This 3D camera is equipped with two cameras trained inwards to produce a rough sense of binocular vision. This provides a good sense of lateral positioning, but a poor understanding of depth. The camera’s depth perception is snapped into focus by a projector placed between the two camera lenses. When firing, the projector emits a bright pattern of blue light that resembles a fine-grain QR code. The camera’s internal computer compares a scaled flat version of this pattern with the one it now sees wrapped across an object’s geometry and measures its deformation across the object’s surface. This brings the camera’s depth accuracy to $\pm 0.5\text{mm}$. Parts are brought to a distance of $\sim 500\text{mm}$ from the camera to sit within its focal length.

The vision system is one of several additions to the robotics cell that enabled this project. Opposite the 3D camera is a Union Graduate cast steel bowl lathe. In this application the lathe was used to index a cutting blade to allow for greater flexibility in cut positioning. The camera and the lathe flank the robot on either side with roughly 4m between them. In front of the robot is a rail system that is hung from the outer wall of the cell. The rails are loaded with branches from outside the cell so the robot can be kept ‘fed’ without the safety perimeter of the cell being broken. A small opening was made in the front wall of the cell with a slide protruding through to allow the finished branch components to exit the cell and be collected. As essential as the vision system is the cutting tool.

- 5. Global diagram illustrating the transition of the forces from the roof structure via the tree column to ground. © Francis Archer, ARUP.
- 6. Highlighted section of the structural system through the Field Station. © Garrett Nelli, Design + Make Postgraduate Programme 2021/2022.
- 7. Robotic gripper and feed station of the branches in their raw state. © Will Gowland.
- 8. Robotic scanning of individual branch geometry. © Will Gowland.
- 9. Production of the repeated branch component of the spaceframe through the use of the indexed circular saw. © Will Gowland.
- 10. Exposing the scanning process to resolve the geometric variety of branch components. © Wyatt Armstrong.



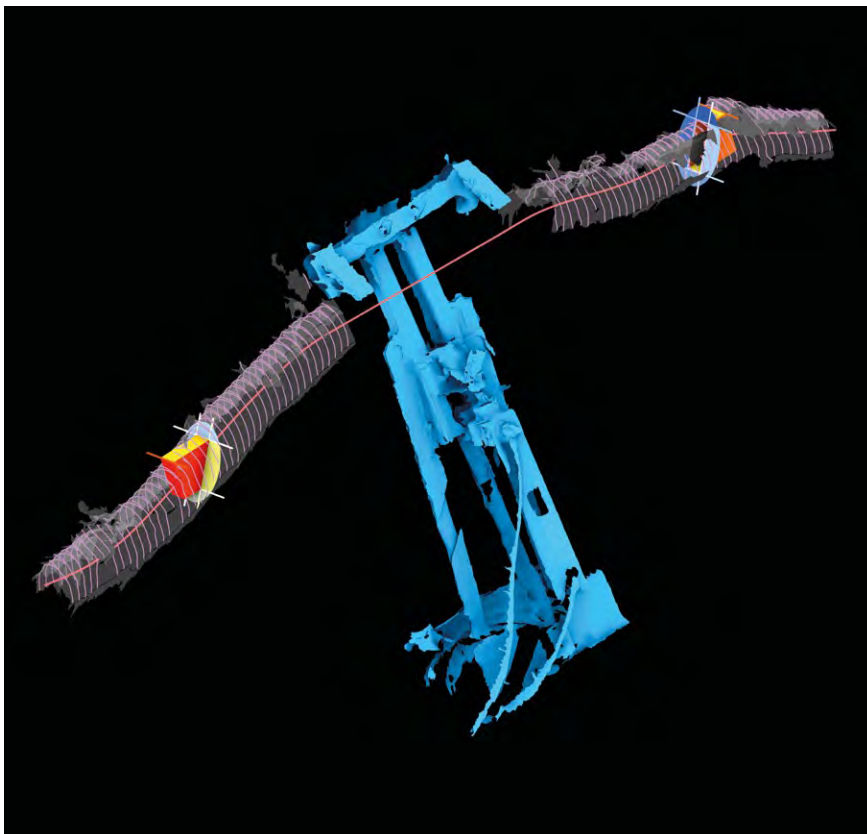
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As branch diameters grew larger through the project, a spindle mounted in a cast steel housing with a 400mm-diameter blade replaced the earlier Makita saw. To carry the branches, the robot is equipped with an in-house-made pneumatic gripper that went through multiple iterations, adapting to increasing branch sizes across the span of the project (Figs. 7, 8, 9).

Using several scripts written in Python and C#, the camera is calibrated and accessed through the environment of a built-in Robot Operating System (ROS). Within the cell, the camera is mounted statically and positionally referenced to the robot's root point. By knowing the camera's coordinate system as an offset from the robot's root point, scans of objects that the robot presents to the camera can be oriented and stitched together by reading the tool control point data at the position from which a scan is taken (Fig. 10).

While the camera is accessed by ROS, requests for the camera to fire are sent from Grasshopper through COMPAS, an open-source computational framework developed by COMPAS Association for Multi-Disciplinary Research in Architecture, Engineering, Fabrication, and Construction. The COMPAS plugin for Grasshopper provides a networked link between ROS topics and information in Grasshopper. Once fired, a point cloud is delivered from the camera to ROS. It is then meshed within ROS and sent over the network to Grasshopper with a timestamp.

Once in Grasshopper, the meshes are linked through their timestamps and compiled into branches composed of four separate scans. The four scans allow roughly three-quarters of the geometry at the branch ends to be documented. Through a contour analysis, the backside of the branch can be estimated by best-fit circles generated by three points on the contours. From the estimated circles, a centreline down the branch is found, to which the modelled geometry of a standardised component can be oriented. In orienting joinery to the branch centrelines, several stages of checks are in place to ensure key geometry of the tenon joints are fully located inside the branch. These involve projecting points on the tenon from the initial orientation back to the branch scan and adjusting accordingly.

The robotics toolpaths are derived from the planes and cut edges of the modelled joinery. Once the modelled joinery is verified to be within the scanned branch, the robotic movements are automatically updated. The motion planning involves matching the planes and defined edges of the tenon joints to several planes located on the circular saw. There are 11 surfaces that define the two tenons at either side of the branch. To prevent the robot's axes from reaching high rotational values, the indexing of the saw blade allows the robot to make cuts perpendicular to each other without much change in position.

The robotic application for the Field Station nests into the academic work of other institutions exploring the analysis of timber through scanning technologies and the robotic fabrication of complex trussed structures. The work of the Field Station has parallels with the Timber Stories project (Svilans *et al.*, 2022), presented at the Works+Words 2022 biennale in Copenhagen, in the analysis of the internal structures of timber through CT scanning for optimised milling layouts. Also analogous is the work at ETH with Gramazio Kohler on House 41781 (Gramazio Kohler Research, 2016), in which the cutting and assembly of standardised timber elements within a spatial trussed assembly was automated. Occupying a territory between these projects, the Field Station does not seek to fully automate the production and assembly of the truss, or to fully comprehend the precise geometry of the pith and branches within the timber, as the continuity of grain is not being severed along the element's length. Instead, it pushes a middle ground between these projects that leverages the inherent strength of round timber and elevates the connection details of what would otherwise be considered a waste product, to furniture-grade levels of craft.



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11. Layout of the assorted components of the Field Station, flat-packed and ready for deployment on site. © Design + Make Postgraduate Programme, Hooke Park.

12. Roundwood and machined timber stitched together and pretensioned, ready for lifting in place. © Design + Make Postgraduate Programme, Hooke Park.

13. The intricacies of a complex assembly logic. © Design + Make Postgraduate Programme, Hooke Park.



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Project credits

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14. The 4-ton roof structure is airlifted in to place and securely fastened to the Beech tree fork columns. © Design + Make Postgraduate Programme, Hooke Park.